

YOGESHKUMAR R

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PROFESSIONAL SUMMARY

Generative AI Software Engineer with 2+ years of experience designing and implementing production-ready Large Language Model (LLM), Retrieval-Augmented Generation (RAG), and GraphRAG solutions for enterprise clients in healthcare, insurance, and research domains. Skilled in translating business requirements into scalable GenAI system architectures using Python, FastAPI, LangChain, and cloud-native AWS and GCP services. Experienced in cross-functional collaboration, OAuth 2.0 / OIDC-based security middleware, and delivering impactful AI software solutions end-to-end.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering 2019 – 2023
Rajalakshmi Engineering College, Anna University *Chennai, Tamil Nadu* GPA: 8.26 / 10.0

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **AI / GenAI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), GraphRAG, Agentic AI, Multi-Agent Systems, Prompt Engineering, Fine-tuning, NLP, Knowledge Graphs, Transformers, Embeddings, Semantic Search, Function Calling, Chain-of-Thought Prompting
- **Frameworks and APIs:** LangChain, LangGraph, LangSmith, FastAPI, Flask, REST APIs, JSON APIs, HuggingFace Transformers, PyTorch, TensorFlow, Scikit-Learn
- **Security:** OAuth 2.0, OpenID Connect (OIDC), JWT, Azure SSO, AWS ALB OIDC, Authentication Middleware, User Audit Logging
- **LLM Platforms:** OpenAI API (GPT-4), Anthropic Claude API, Google Gemini, AWS Bedrock, GCP Vertex AI, OpenRouter
- **Cloud / DevOps:** AWS (Bedrock, ECS Fargate, S3, CodePipeline, DocumentDB, OpenSearch), GCP (Vertex AI, VM), Azure, Pinecone, FAISS, Neo4j, MongoDB, MySQL, Docker, Git, CI/CD

EXPERIENCE

Generative AI Engineer Sep 2023 – Present
Virtusa *Chennai, Tamil Nadu*

- Designed and implemented GenAI software solutions for 3+ enterprise clients, translating business requirements into technical specifications for scalable RAG and GraphRAG pipelines using Python, LangChain, and Neo4j.
- Built and containerized scalable REST/JSON API services using FastAPI and Docker deployed on AWS ECS/Fargate, enabling low-latency LLM-powered inference pipelines for enterprise applications.
- Implemented OAuth 2.0 / OIDC authentication middleware using Azure SSO and AWS ALB OIDC headers; extracted and encrypted JWT tokens for user identity validation and audit logging across GenAI services.
- Developed semantic search and retrieval pipelines using FAISS, Pinecone, and AWS OpenSearch; applied advanced prompt engineering (few-shot, chain-of-thought, RAG) to reduce hallucinations.
- Optimized chunking strategies, embedding configurations, and retrieval metrics (precision, recall, MRR), improving retrieval quality by 50%.
- Collaborated with cross-functional product, ML, and DevOps teams to deliver GenAI initiatives; trained 15+ engineers on LangChain, RAG architectures, and LLM evaluation techniques.

Project Intern May 2022 – Mar 2023
Virtusa *Chennai, Tamil Nadu*

- Built an ML-based Loan Approval Prediction Dashboard using Python, Flask, and Scikit-Learn, enabling non-technical users to train and evaluate classification models via a web UI.

PROJECTS

GraphRAG Test Case Generation – AstraZeneca *Virtusa — Client Project*

- Architected an end-to-end GraphRAG pipeline using Python, LangChain, and Neo4j to auto-generate enterprise test cases from requirement documents, reducing manual authoring effort by 40%; integrated Azure SSO for secure access and deployed full AWS stack (Bedrock, OpenSearch, ECS Fargate, CodePipeline) with CI/CD automation.

RAG-Based Test Case Generation – Reliance Standard Life Insurance *Virtusa — Client Project*

- Built end-to-end RAG pipelines using Python, LangChain, GCP Vertex AI, and Anthropic Claude to auto-generate test cases from 500+ business requirements; deployed containerized REST API services on GCP with MongoDB and implemented full LLM output validation.

Impact-based Test Case Selection & Recommendation

Virtusa — Internal POC

- Designed an AI-powered test impact analysis system using GitHub API, LangChain, Google Gemini, few-shot prompting, and Pinecone to map developer code diffs to affected test cases, eliminating full regression cycles and enabling targeted retesting of impacted features only.

CERTIFICATIONS

AWS Certified AI Practitioner – Foundational	<i>October 2025</i>
ISTQB Certified Tester Foundation Level (CTFL)	<i>January 2025</i>
OCI Generative AI Professional	<i>June 2024</i>
AWS Certified Generative AI Professional	<i>(In Progress)</i>

AWARDS

Jatayu Explorer Award – Runner-Up, Virtusa – Recognized for impactful ML-based automation solution design.

AI for Bharat Hackathon – Prototype Phase Qualifier – Selected among top teams to advance to the Prototype phase out of national-level competition; results pending (April 2026).